THE FREE HEYTING CATEGORY ON A PRETOPOS IS A PRETOPOS ONLY IN THE BOOLEAN CASE: THE INTUITIONISTIC TRANSLATION AND THE PRETOPOS COMPLETION

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Abstract. Reading a coherent theory in intuitionistic first-order logic is the free Heyting category $\mathsf{H}\mathcal{C}$ on its coherent syntactic category $\mathcal{C}$; adding the imaginaries is the pretopos completion P. We prove that the two operations commute only trivially: the canonical functor $\mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ is always full and faithful, and it is an equivalence if and only if $\mathcal{C}$ is Boolean, in which case H does nothing. The objects we show to be missing from $\mathsf{HP}\mathcal{C}$ are two copies of an object glued along the intuitionistic negation of a non-complemented subobject; they are excluded by an invariant of the objects of $\mathsf{HP}\mathcal{C}$ in Kripke models over a fork, whose existence for every non-Boolean $\mathcal{C}$ is a form of the homomorphism preservation theorem. H commutes with the positive completion; with the exact completion it commutes for propositional theories and fails when some object has two points in the source model of a homomorphism crossing a subobject. For subobjects there is a preservation theorem: an intuitionistic formula is coherent if and only if its forcing is reflected along the transitions which are isomorphisms of models; so the equivalence relations on coherent imaginaries with a quotient in $\mathsf{HP}\mathcal{C}$ are exactly the coherent ones. For finite propositional theories the image of the comparison is characterised by a condition on zigzags in one finite Kripke model. The Morleyization being H followed by excluded middle, which commutes with P, its Galois criterion says when excluded middle repairs the comparison.

1. Introduction

A coherent theory T has a coherent syntactic category $\mathcal{C}_T$, and three constructions on it are the subject of this paper and of its predecessor [5]. The pretopos completion $\mathsf{P}\mathcal{C}$ adds finite coproducts of sorts and quotients by definable equivalence relations, the coherent imaginaries [1, A3.3, D1.4]. The Morleyization $\mathsf{B}\mathcal{C}$ is the syntactic category of T read in classical first-order logic. Between them lies the Heyting completion $\mathsf{H}\mathcal{C}$, the syntactic

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category of T read in intuitionistic first-order logic, and $\mathsf{B} \simeq \mathsf{KH}$ with K the addition of excluded middle [5, §9].

In [5] the order of translation and completion was studied for B: there is a full and faithful comparison $\mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$, which is an equivalence exactly when a Galois criterion holds [5, Thm. 6.1]; it fails for the theory of complete multipartite graphs and holds, for example, for the theory of an equivalence relation. It was also shown that K commutes with P on all Heyting categories [5, Thm. 9.2], so that the comparison for B is the image under K of the comparison

$$c_{\mathcal{C}} \colon \mathsf{HP}\mathcal{C} \longrightarrow \mathsf{PH}\mathcal{C}$$

for H. The question left open was when $c_{\mathcal{C}}$ itself is an equivalence. The answer is the following.

Theorem 1.1. *For a small coherent category $\mathcal{C}$ the functor $c_{\mathcal{C}}$ is full and faithful, and the following are equivalent.*

(i) *$c_{\mathcal{C}} \colon \mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ is an equivalence.*
(ii) *$\mathsf{HP}\mathcal{C}$ is a pretopos.*
(iii) *$\mathcal{C}$ is Boolean.*

If they hold, the units $\mathcal{C} \to \mathsf{H}\mathcal{C}$ and $\mathsf{P}\mathcal{C} \to \mathsf{HP}\mathcal{C}$ are equivalences.

For a pretopos $\mathcal{C}$ this says that the free Heyting category on $\mathcal{C}$ is a pretopos if and only if $\mathcal{C}$ is Boolean (Corollary 5.1).

What is new and what is known. New, to the author's knowledge: Theorem 1.1; the invariant that proves it (Lemma 3.1); the commutation of H with the positive completion (Proposition 6.2) and the resulting location of the failure (Corollary 6.3, Proposition 6.4); the preservation theorem for rigid formulas and the description of the effective equivalence relations (Theorems 7.3 and 7.4); and the reading of the Galois criterion of [5] as the statement that $c_{\mathcal{C}}$ is inverted by K (Corollary 8.1). Known and used: syntactic categories of first-order theories are Heyting categories and classify their models in Heyting categories [1, D1.4]; the pretopos completion [1, A3.3]; completeness of coherent logic for set-valued models [1, D1.5], [4]; the homomorphism preservation theorem of Łoś, Tarski and Lyndon, of which Lemma 4.1 is the form needed here and is proved in full; the Heyting structure of presheaf categories, that is, Kripke semantics, and Joyal's completeness theorem for it [3]; and from [5] the facts that $\mathsf{P}\mathcal{D}$ is Heyting when $\mathcal{D}$ is, with the expected universal property, and that K commutes with P.

Conventions. Coh is the $(2,1)$-category of small coherent categories, coherent functors and natural isomorphisms, and **Heyt** that of small Heyting categories (coherent categories in which every pullback functor $f^* \colon \mathrm{Sub}(B) \to \mathrm{Sub}(A)$ has a right adjoint $\forall_f$), Heyting functors (coherent functors preserving the $\forall_f$) and natural isomorphisms [1, A1.4]. A Heyting functor preserves the implication of subobjects and the pseudo-complement

$\neg U = (U \Rightarrow 0)$, since $U \Rightarrow V = \forall_i(i^*V)$ for the inclusion $i\colon U \to A$. A coherent category is *Boolean* if every subobject is complemented. A model of $\mathcal{C}$ is a coherent functor $\mathcal{C} \to \mathbf{Set}$, and $\mathrm{Mod}(\mathcal{C})$ is the category of models with all natural transformations, the homomorphisms. The universal properties below are stated for small categories but hold, with the same proofs, for targets such as $\mathbf{Set}^{\mathcal{K}}$, $\mathcal{K}$ a small category; alternatively one replaces $\mathbf{Set}^{\mathcal{K}}$ by a small full subcategory closed under the operations of a Heyting pretopos and containing the objects in question.

2. The two completions and the comparison

The Heyting completion. For $\mathcal{C}$ in **Coh** let $\mathsf{H}\mathcal{C}$ be the syntactic category of the internal coherent theory of $\mathcal{C}$ (a sort for every object, a function symbol for every morphism, the coherent sequents valid in $\mathcal{C}$) read in intuitionistic first-order logic with possibly empty sorts (the initial object is a sort), and $\eta\colon \mathcal{C} \to \mathsf{H}\mathcal{C}$ the coherent functor sending an object to its sort. $\mathsf{H}\mathcal{C}$ is a Heyting category, and for every Heyting category $\mathcal{D}$ restriction along η is an equivalence of groupoids

$$\mathbf{Heyt}(\mathsf{H}\mathcal{C}, \mathcal{D}) \;\simeq\; \mathbf{Coh}(\mathcal{C}, \mathcal{D})\,:$$

a coherent functor $\mathcal{C} \to \mathcal{D}$ is a model of the internal theory in $\mathcal{D}$, it interprets every first-order formula, intuitionistic logic is sound in Heyting categories, and an isomorphism of models preserves the interpretations of all formulas [1, D1.4], [5, §9]. Every object of $\mathsf{H}\mathcal{C}$ is a subobject of some ηA: it is a formula in a context $x_1{:}A_1, \ldots, x_n{:}A_n$, a subobject of $\eta(A_1 \times \cdots \times A_n)$.

The forgetful 2-functor $\mathbf{Heyt} \to \mathbf{Coh}$ is not full, a coherent functor between Heyting categories need not be Heyting. So H is a free construction and not a reflector; the join lemma of [6, 5], which concerns reflectors, does not apply to it, and we compare the composites HP and PH directly.

Lemma 2.1. *If $\mathcal{C}$ is Boolean, then $\eta\colon \mathcal{C} \to \mathsf{H}\mathcal{C}$ is an equivalence.*

Proof. A Boolean coherent category is Heyting, with $\forall_f U = \neg\exists_f\neg U$. Let $G\colon \mathcal{C} \to \mathcal{D}$ be coherent with $\mathcal{D}$ Heyting. Then G preserves complements and images, so $G(\forall_f U)$ is a complement W of $\exists_{Gf}V'$, where $V' = G(\neg U)$ is a complement of $V = GU$. In a Heyting category this W is $\forall_{Gf}V$: for a subobject W', $(Gf)^*W' \le V$ iff $(Gf)^*W' \wedge V' = 0$ iff $W' \wedge \exists_{Gf}V' = 0$ (Frobenius reciprocity) iff $W' \le W$. So every coherent functor from $\mathcal{C}$ to a Heyting category is Heyting, and the identity of $\mathcal{C}$ has the universal property of η. □

The pretopos completion of a Heyting category. We write $\iota\colon \mathcal{C} \to \mathsf{P}\mathcal{C}$ for the unit of the pretopos completion; it is full and faithful, and restriction along it is an equivalence between the categories of coherent functors $\mathsf{P}\mathcal{C} \to \mathcal{E}$ and $\mathcal{C} \to \mathcal{E}$ with all natural transformations (not only the groupoids $\mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{E})$ and $\mathbf{Coh}(\mathcal{C}, \mathcal{E})$), when $\mathcal{E}$ is a pretopos [1, A3.3]. In particular $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$. Every object of $\mathsf{P}\mathcal{C}$ is the quotient of a finite coproduct of objects ιA by an equivalence relation.

Lemma 2.2 ([5, Lemma 9.1])**.** *If $\mathcal{D}$ is a Heyting category, then $\mathsf{P}\mathcal{D}$ is a Heyting pretopos and $\iota\colon \mathcal{D} \to \mathsf{P}\mathcal{D}$ is a Heyting functor; and for a Heyting pretopos $\mathcal{E}$ a coherent functor $\mathsf{P}\mathcal{D} \to \mathcal{E}$ is Heyting if and only if its restriction to $\mathcal{D}$ is.*

Consequently $\mathsf{PH}\mathcal{C}$ is the free Heyting pretopos on $\mathcal{C}$: $\mathbf{Heyt}(\mathsf{PH}\mathcal{C}, \mathcal{E}) \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{E})$ for Heyting pretoposes $\mathcal{E}$.

The comparison. The coherent functor $\mathsf{P}\eta\colon \mathsf{P}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ has a Heyting target, so it extends, uniquely up to a unique isomorphism, to a Heyting functor

$$\begin{array}{ccc} \mathsf{P}\mathcal{C} & \xrightarrow{\eta_{\mathsf{P}\mathcal{C}}} & \mathsf{HP}\mathcal{C} \\ & \searrow^{\mathsf{P}\eta} & \downarrow{\scriptstyle c_{\mathcal{C}}} \\ & & \mathsf{PH}\mathcal{C}. \end{array}$$

Lemma 2.3. *$c_{\mathcal{C}}$ is full and faithful. It is an equivalence if and only if $\mathsf{HP}\mathcal{C}$ is a pretopos.*

Proof. For a Heyting pretopos $\mathcal{E}$, by Lemma 2.2 and the universal properties,

$$\begin{aligned} \mathbf{Heyt}(\mathsf{PHP}\mathcal{C}, \mathcal{E}) &\simeq \mathbf{Heyt}(\mathsf{HP}\mathcal{C}, \mathcal{E}) \simeq \mathbf{Coh}(\mathsf{P}\mathcal{C}, \mathcal{E}) \\ &\simeq \mathbf{Coh}(\mathcal{C}, \mathcal{E}) \simeq \mathbf{Heyt}(\mathsf{PH}\mathcal{C}, \mathcal{E}), \end{aligned}$$

by restriction to $\mathcal{C}$. So there is a Heyting equivalence $e\colon \mathsf{PHP}\mathcal{C} \simeq \mathsf{PH}\mathcal{C}$ compatible with the functors from $\mathcal{C}$, and $e \circ \iota_{\mathsf{HP}\mathcal{C}}$ is a Heyting functor $\mathsf{HP}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ whose restriction to $\mathsf{P}\mathcal{C}$ is a coherent functor into a pretopos restricting on $\mathcal{C}$ as $\mathsf{P}\eta$ does; hence $e \circ \iota_{\mathsf{HP}\mathcal{C}} \cong c_{\mathcal{C}}$. The unit $\iota_{\mathsf{HP}\mathcal{C}}$ is full and faithful, and it is an equivalence if and only if $\mathsf{HP}\mathcal{C}$ is a pretopos. □

3. Kripke models and an invariant

Let $\mathcal{K}$ be a small category. The category $\mathbf{Set}^{\mathcal{K}}$ is a Heyting pretopos, and limits, images, unions, coproducts and quotients in it are computed at each object of $\mathcal{K}$; so a coherent functor $\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}$ is the same as a functor $M\colon \mathcal{K} \to \mathrm{Mod}(\mathcal{C})$, a *Kripke model* of $\mathcal{C}$ over $\mathcal{K}$; the transitions are arbitrary homomorphisms and need not be injective. We write M_k for the model at k. The pseudo-complement in $\mathbf{Set}^{\mathcal{K}}$ is the Kripke negation: for a subfunctor $S \subseteq A$,

$$(\neg S)(k) = \{x \in A(k) \mid A(w)x \notin S(l) \text{ for every arrow } w\colon k \to l\}, \tag{1}$$

since the right-hand side is a subfunctor disjoint from S, and every subfunctor W disjoint from S lies in it ($x \in W(k)$ gives $A(w)x \in W(l)$).

A Kripke model M extends to

$$\begin{aligned} M^{\mathsf{H}}&\colon \mathsf{H}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, & M^{\mathsf{P}}&\colon \mathsf{P}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, \\ M^{\mathsf{HP}}&\colon \mathsf{HP}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, & M^{\mathsf{PH}}&\colon \mathsf{PH}\mathcal{C} \to \mathbf{Set}^{\mathcal{K}}, \end{aligned}$$

the first, third and fourth Heyting (Lemma 2.2), with $M^{\mathsf{HP}}\eta_{\mathsf{P}\mathcal{C}} \cong M^{\mathsf{P}}$ and

$$M^{\mathsf{PH}} \circ c_{\mathcal{C}} \;\cong\; M^{\mathsf{HP}}, \tag{2}$$

both sides being Heyting functors whose restrictions to $\mathsf{P}\mathcal{C}$ are coherent functors into a pretopos restricting to M on $\mathcal{C}$. The functor M^{P} is the Kripke model of $\mathsf{P}\mathcal{C}$ corresponding to M under $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$: for Y in $\mathsf{P}\mathcal{C}$, the set $M^{\mathsf{P}}(Y)(k)$ is the value at Y of the extension of M_k, and an arrow w of $\mathcal{K}$ acts by the extension of the homomorphism $M(w)$.

Lemma 3.1 (invariant)**.** *Let X be an object of $\mathsf{HP}\mathcal{C}$, M a Kripke model of $\mathcal{C}$ over $\mathcal{K}$, and $w\colon k \to l$ an arrow of $\mathcal{K}$ such that $M(w)$ is an isomorphism of models. Then $M^{\mathsf{HP}}(X)(w)$ is injective.*

Proof. X is a subobject of $\eta_{\mathsf{P}\mathcal{C}}Y$ for an object Y of $\mathsf{P}\mathcal{C}$, so $M^{\mathsf{HP}}(X)$ is a subfunctor of $M^{\mathsf{P}}(Y)$. The functor $\mathcal{K} \to \mathrm{Mod}(\mathsf{P}\mathcal{C})$ corresponding to M^{P} sends w to an isomorphism, since $M(w)$ is one and $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$; so $M^{\mathsf{P}}(Y)(w)$ is a bijection and its restriction is injective. □

When $\mathcal{C}$ is not Boolean, some objects of $\mathsf{PH}\mathcal{C}$ do not have this property (Section 5): they are quotients by relations whose truth at k depends on all the arrows out of k and not only on the model M_k.

4. Non-Boolean categories have crossing homomorphisms

Lemma 4.1. *Let $S \leq A$ be a subobject in a small coherent category $\mathcal{C}$ which is not complemented. Then there are models M, N of $\mathcal{C}$, a homomorphism $h\colon M \to N$ and $a \in A(M) \setminus S(M)$ with $h_A(a) \in S(N)$.*

Proof. For $U \leq B$ with inclusion i we write $U(x)$ for the formula $\exists u{:}U\,(i(u) = x)$. We use the completeness theorem: if $U, U_1, \ldots, U_n \leq B$ and $U(M) \subseteq U_1(M) \cup \cdots \cup U_n(M)$ for all models M, then $U \leq U_1 \vee \cdots \vee U_n$ [1, D1.5], [4]; and the compactness theorem for the classical first-order theory $T_{\mathcal{C}}$ whose models are the models of $\mathcal{C}$, in many-sorted logic with possibly empty sorts (the initial object is a sort).

Let $I = \{U \leq A \mid U \wedge S = 0\}$, an ideal of the distributive lattice $\mathrm{Sub}(A)$, and J the ideal of the W with $W \leq S \vee U$ for some $U \in I$. It is proper: $A \leq S \vee U$ with $U \wedge S = 0$ would make U a complement of S. By the prime filter theorem there is a prime filter $\mathfrak{p}$ of $\mathrm{Sub}(A)$ disjoint from J.

There are a model M and $a \in A(M)$ with $a \in U(M)$ iff $U \in \mathfrak{p}$, for all $U \leq A$: the theory $T_{\mathcal{C}}$ with a new constant c of sort A and the axioms $U(c)$, $U \in \mathfrak{p}$, and $\neg U(c)$, $U \notin \mathfrak{p}$, is finitely satisfiable, since otherwise there are $U_0 \in \mathfrak{p}$ and $U_1, \ldots, U_n \notin \mathfrak{p}$ with $U_0(M) \subseteq \bigcup_i U_i(M)$ in all models, hence $U_0 \leq \bigvee_i U_i$, and $\bigvee_i U_i \in \mathfrak{p}$ contradicts the primeness of $\mathfrak{p}$ (for $n = 0$, $0 \in \mathfrak{p}$ contradicts its properness). In particular $a \notin S(M)$, and $a \notin U(M)$ for $U \in I$.

Now add to $T_{\mathcal{C}}$ a constant c_b of sort B for every object B and every $b \in B(M)$, the axioms

$$f(c_b) = c_{f_M(b)} \qquad (f\colon B \to B' \text{ a morphism of } \mathcal{C},\ b \in B(M)),$$

and the axiom $S(c_a)$. A finite part of these axioms mentions, besides c_a, finitely many distinct constants $c_{b_1}, \ldots, c_{b_n}$, $b_i \in B_i(M)$, and finitely many

equations between them and c_a. Put $b_0 = a$, $B_0 = A$ and $B = \prod_{i=0}^{n} B_i$, and let $\theta \leq B$ be the subobject defined by the conjunction of these equations, one variable y_i replacing each constant c_{b_i} (an equation $f(y_i) = y_j$ defines an equaliser, a subobject of B); then $(a, b_1, \ldots, b_n) \in \theta(M)$. If the finite part had no model, every model N' would satisfy $\exists_{\pi_0}\theta(N') \cap S(N') = \emptyset$, so $U := \exists_{\pi_0}\theta \leq A$ would satisfy $U \wedge S = 0$ by completeness, that is $U \in I \subseteq J$; but $a \in U(M)$ gives $U \in \mathfrak{p}$, a contradiction. By compactness there is a model N of all these axioms, and $h_B(b) := c_b^N$ is, by the axioms, a natural transformation $M \to N$, with $h_A(a) \in S(N)$. □

Remark 4.2. Lemma 4.1 is the homomorphism preservation theorem in the form: a subobject whose complement is preserved by all homomorphisms of models is complemented. Its converse is trivial, so a coherent category is Boolean if and only if every homomorphism of its models reflects every subobject. In that case every first-order formula is equivalent to a coherent one and the homomorphisms are the elementary embeddings of the models of the corresponding classical theory. (A single homomorphism reflecting all subobjects need not be elementary: the inclusion of a two-element set in a three-element set, for pure equality.)

5. Proof of the main theorem

Proof of Theorem 1.1. The first assertion and (i)⇔(ii) are Lemma 2.3.

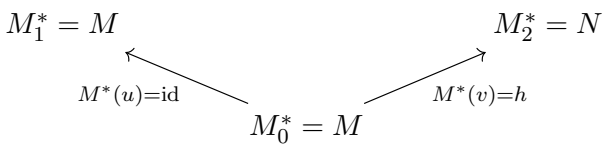
(iii)⇒(i). If $\mathcal{C}$ is Boolean, so is $\mathsf{P}\mathcal{C}$ [5, Prop. 3.4]: complements are taken componentwise in finite coproducts, and the complement of a saturated subobject is saturated. By Lemma 2.1, $\eta_{\mathcal{C}}$ and $\eta_{\mathsf{P}\mathcal{C}}$ are equivalences; so $\mathsf{HP}\mathcal{C} \simeq \mathsf{P}\mathcal{C}$ is a pretopos.

(i)⇒(iii). Suppose that $S \leq A$ is not complemented in $\mathcal{C}$, and let $h\colon M \to N$ and a be as in Lemma 4.1. In $\mathsf{PH}\mathcal{C}$ let $\neg S \leq A$ be the pseudo-complement of (the image of) S in (the image of) A, and let

$$Q \;:=\; (A + A)/E, \qquad E := \Delta_{A+A} \vee (\kappa_1, \kappa_2)(\neg S) \vee (\kappa_2, \kappa_1)(\neg S),$$

with $\kappa_1, \kappa_2\colon A \to A + A$ the coprojections and $(\kappa_i, \kappa_j)(\neg S)$ the image of $\neg S$ in $(A + A) \times (A + A)$: the two copies of A glued along $\neg S$, the pushout of $A \leftarrow \neg S \to A$. The relation E is an equivalence relation, so Q exists in the pretopos $\mathsf{PH}\mathcal{C}$. We show that Q is not isomorphic to $c_{\mathcal{C}}(X)$ for any object X of $\mathsf{HP}\mathcal{C}$.

Let $\mathcal{K}$ be the fork, the poset with three elements $0 < 1$ and $0 < 2$, with arrows $u\colon 0 \to 1$ and $v\colon 0 \to 2$, and M^* the Kripke model



Since $M^{*\mathsf{PH}}$ is a Heyting functor between pretoposes, it preserves pseudo-complements, coproducts, images and quotients; so $M^{*\mathsf{PH}}(Q)$ is the quotient

of $M^*(A) + M^*(A)$ which identifies the two copies of $x \in A(M_k^*)$ exactly when $x \in (\neg M^*(S))(k)$. By (1), $a \in (\neg M^*(S))(1)$, because the only arrow out of 1 is the identity and $a \notin S(M)$; and $a \notin (\neg M^*(S))(0)$, because $M^*(A)(v)a = h_A(a) \in S(N)$. Hence the two copies of a are distinct points of $M^{*\mathsf{PH}}(Q)(0)$ with the same image under u in $M^{*\mathsf{PH}}(Q)(1)$.

If $Q \cong c_{\mathcal{C}}(X)$, then by (2) $M^{*\mathsf{PH}}(Q) \cong M^{*\mathsf{HP}}(X)$, whose map along u is injective by Lemma 3.1, since $M^*(u)$ is an isomorphism. This is a contradiction; so $c_{\mathcal{C}}$ is not essentially surjective.

The last assertion is Lemma 2.1 for $\mathcal{C}$ and $\mathsf{P}\mathcal{C}$. □

Corollary 5.1. *Let $\mathcal{C}$ be a small pretopos. The free Heyting category $\mathsf{H}\mathcal{C}$ on $\mathcal{C}$ is a pretopos if and only if $\mathcal{C}$ is Boolean, and then $\mathsf{H}\mathcal{C} \simeq \mathcal{C}$.*

Proof. $\iota\colon \mathcal{C} \to \mathsf{P}\mathcal{C}$ is an equivalence, so $\mathsf{H}\mathcal{C} \simeq \mathsf{HP}\mathcal{C}$; apply Theorem 1.1. □

Example 5.2 (pure equality)**.** Let $\mathcal{C}$ be the syntactic category of the coherent theory with one sort σ and no symbols and axioms. The diagonal of $\sigma \times \sigma$ is not complemented; a crossing homomorphism is the map from a two-element set $\{a, b\}$ to a one-element set, with the pair (a, b). The object Q is two copies of $\sigma \times \sigma$ glued along $\neg(x = y)$: an intuitionistic imaginary of the theory of pure equality which is not an intuitionistically definable subset of a coherent imaginary. Over the fork, $\neg(x = y)$ is empty at 0 and is the set of the two off-diagonal pairs at 1; so the fibre of Q at 0 has eight points, two copies of the four pairs, and the fibre at 1 has six, the two copies of (a, b) and of (b, a) being identified.

Example 5.3 (propositional theories)**.** Let $\mathcal{C} = D$ be a bounded distributive lattice, a propositional coherent theory. Up to isomorphism its models are the prime filters and the homomorphisms are the inclusions, and Lemma 4.1 is the classical fact that a distributive lattice in which the prime filters form an antichain is Boolean. $\mathsf{H}D$ is the free Heyting algebra on the distributive lattice D. For the three-element chain $0 < p < 1$ the object Q is $(1 + 1)/E$ with the two points identified when $\neg p$; over the fork $\{1\} = \{1\} \subset \{p, 1\}$ of prime filters it has two points at 0 and one at 1, and no intuitionistically definable subobject of an object of $\mathsf{P}D$ behaves so.

6. Where the failure is: coproducts and quotients together

The pretopos completion is $\mathsf{P} = \mathsf{EF}$, with F the positive completion (finite disjoint coproducts) and E the exact completion of a regular category [1, A3.3], [6]. Here a coherent category is *positive* if it has finite coproducts, disjoint and stable under pullback, F is the reflector onto the positive coherent categories (coherent functors from $\mathsf{F}\mathcal{C}$ to a positive coherent category correspond to those from $\mathcal{C}$; the unit is full and faithful and full on subobjects, and every object of $\mathsf{F}\mathcal{C}$ is a finite coproduct of objects of $\mathcal{C}$), and E is the exact (ex/reg) completion of a regular category. The object Q uses a coproduct and a quotient. The coproduct step alone is never responsible;

the quotient step alone is not responsible for propositional theories, and is responsible under the hypothesis of Proposition 6.4.

Lemma 6.1. *If $\mathcal{D}$ is a Heyting category, then $\mathsf{F}\mathcal{D}$ is a Heyting category and the unit $\mathcal{D} \to \mathsf{F}\mathcal{D}$ is Heyting; and for a positive Heyting category $\mathcal{D}'$ a coherent functor $\mathsf{F}\mathcal{D} \to \mathcal{D}'$ is Heyting if and only if its restriction to $\mathcal{D}$ is.*

Proof. Every object of $\mathsf{F}\mathcal{D}$ is a finite coproduct $\coprod_i A_i$ of objects of $\mathcal{D}$, the unit is full and faithful and full on subobjects, and $\mathrm{Sub}(\coprod_i A_i) \cong \prod_i \mathrm{Sub}_{\mathcal{D}}(A_i)$, $U \mapsto (U \wedge A_i)_i$. Let $f\colon \coprod_i A_i \to \coprod_j B_j$, and let $A_{ij} = A_i \wedge f^* B_j$, a subobject of A_i in $\mathcal{D}$, and $f_{ij}\colon A_{ij} \to B_j$ the restriction, a morphism of $\mathcal{D}$; for each i the A_{ij} are disjoint with union A_i. For $V = (V_j)_j$ one has $f^*V \wedge A_i = \bigvee_j f_{ij}^* V_j$, and therefore, for $U \leq \coprod_i A_i$,

$$(\forall_f U) \wedge B_j \;=\; \bigwedge_i \forall_{f_{ij}} (U \wedge A_{ij}) :$$

$V \leq \forall_f U$ iff $f_{ij}^* V_j \leq U \wedge A_{ij}$ for all i, j iff $\bigvee_j f_{ij}^* V_j \leq U \wedge A_i$ for all i, the A_{ij} being disjoint. The formula involves only the $\forall$ of $\mathcal{D}$, finite meets and the decomposition of coproducts; so the unit preserves $\forall$, and so does a coherent functor to a positive Heyting category whose restriction to $\mathcal{D}$ is Heyting. The converse holds since the unit is Heyting. □

Proposition 6.2. *For every small coherent category $\mathcal{C}$ the canonical Heyting functor $\mathsf{HF}\mathcal{C} \to \mathsf{FH}\mathcal{C}$ is an equivalence.*

Proof. By Lemma 6.1, $\mathsf{FH}\mathcal{C}$ is a positive Heyting category with $\mathbf{Heyt}(\mathsf{FH}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Heyt}(\mathsf{H}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{D}')$ for positive Heyting categories $\mathcal{D}'$. On the other side $\mathbf{Heyt}(\mathsf{HF}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathsf{F}\mathcal{C}, \mathcal{D}') \simeq \mathbf{Coh}(\mathcal{C}, \mathcal{D}')$. So it suffices to show that $\mathsf{HF}\mathcal{C}$ is positive; then both categories are the free positive Heyting category on $\mathcal{C}$, and the canonical functor, which is compatible with the functors from $\mathcal{C}$, is an equivalence. Let $X \leq \eta Y$ and $X' \leq \eta Y'$ be objects of $\mathsf{HF}\mathcal{C}$, with Y, Y' in $\mathsf{F}\mathcal{C}$. The coherent functor η sends the coproduct $Y + Y'$, the union of the disjoint subobjects Y and Y', to a union of disjoint subobjects, which in a coherent category is a coproduct, disjoint and stable [1, A1.4]; so $X \vee X' \leq \eta(Y + Y')$ is a disjoint and stable coproduct of X and X', and the initial object is the empty one. □

Corollary 6.3. *Let $\mathcal{C}$ be a small positive coherent category. Then $\mathsf{HE}\mathcal{C}$ is positive, and it is exact if and only if $\mathcal{C}$ is Boolean. For a small bounded distributive lattice D, on the other hand, $\mathsf{E}D \simeq D$ and $\mathsf{HE}D \simeq \mathsf{H}D$ is exact, although $\mathsf{HP}D$ is a pretopos only if D is Boolean.*

Proof. For positive $\mathcal{C}$ the category $\mathsf{E}\mathcal{C}$ is a pretopos, equivalent to $\mathsf{P}\mathcal{C}$ and to $\mathsf{FE}\mathcal{C}$; so $\mathsf{HE}\mathcal{C} \simeq \mathsf{HFE}\mathcal{C}$ is positive by the proof of Proposition 6.2, and a positive coherent category is a pretopos iff it is exact; Theorem 1.1 applies. In a preorder every equivalence relation on an object U is the whole of $U \times U \cong U$, the kernel pair of the identity; so D and the Heyting algebra $\mathsf{H}D$ are exact. □

For general coherent categories the quotient step alone does obstruct, as soon as a model at which a subobject is crossed has two points. For a coherent category $\mathcal{C}$ the exact completion $\mathsf{E}\mathcal{C}$ is coherent and effective (every equivalence relation is a kernel pair; a *cover* is a regular epimorphism), and restriction along the unit is an equivalence between the coherent functors from $\mathsf{E}\mathcal{C}$ and from $\mathcal{C}$ to an effective coherent category, with all natural transformations [6]; in particular $\mathrm{Mod}(\mathsf{E}\mathcal{C}) \simeq \mathrm{Mod}(\mathcal{C})$, and Lemma 3.1 holds for $\mathsf{HE}\mathcal{C}$ with the same proof: for an object X of $\mathsf{HE}\mathcal{C}$ and a Kripke model M of $\mathcal{C}$, the extension of M to $\mathsf{HE}\mathcal{C}$ sends X to a functor whose maps along the arrows w with $M(w)$ an isomorphism are injective.

Proposition 6.4. *Let $\mathcal{C}$ be a small coherent category, $S \leq A$ a subobject, $h\colon M \to N$ a homomorphism of models and $a \in A(M) \setminus S(M)$ with $h_A(a) \in S(N)$. If some object B of $\mathcal{C}$ has at least two points in M, then $\mathsf{HE}\mathcal{C}$ is not effective: the equivalence relation*

$$E\big((x,y),(x',y')\big) \;:=\; (x = x' \wedge y = y') \;\vee\; \big(x = x' \wedge \neg S(x)\big)$$

on $A \times B$ in $\mathsf{HE}\mathcal{C}$, which collapses the fibre over x when $\neg S(x)$, is not a kernel pair.

Proof. E is reflexive and symmetric, and transitive because each disjunct contains $x = x'$ and the second depends only on x. Suppose that E is the kernel pair of a morphism of $\mathsf{HE}\mathcal{C}$; replacing it by the cover in its image factorisation, E is the kernel pair of a cover $q\colon A \times B \to X$. Let M^* be the Kripke model over the fork of the proof of Theorem 1.1 and G its extension to $\mathsf{HE}\mathcal{C}$, a Heyting functor to $\mathbf{Set}^{\mathcal{K}}$. It preserves kernel pairs, covers and pseudo-complements, so $G(X)(k)$ is the quotient of $A(M^*_k) \times B(M^*_k)$ by the relation identifying (x,y) and (x,y') when $x \in (\neg M^*(S))(k)$. As in that proof, $a \notin (\neg M^*(S))(0)$ and $a \in (\neg M^*(S))(1)$; so for $y \neq y'$ in $B(M)$ the points (a,y) and (a,y') are distinct in $G(X)(0)$ and have the same image in $G(X)(1)$. This contradicts the invariant, $M^*(u)$ being an isomorphism. □

Example 6.5. For the theory of pure equality (Example 5.2) take $A = \sigma \times \sigma$, S the diagonal, the map from a two-element set to a one-element set, and $B = \sigma$: $\mathsf{HE}\mathcal{C}$ is not effective, so the Heyting completion does not commute with the exact completion, without any coproduct. In a bounded distributive lattice every object has at most one point in every model, and Proposition 6.4 does not apply, in accordance with Corollary 6.3.

So the Heyting completion commutes with adding coproducts; with adding quotients it commutes for propositional theories and does not commute when a subobject is crossed at a model in which some object has two points. Whether it commutes with adding quotients in the remaining cases, those of the non-Boolean categories in which every model at which a subobject is crossed gives at most one point to every object, is open (Question 9.1).

7. Rigid formulas are coherent, and the effective equivalence relations

The invariant of Lemma 3.1, injectivity of the transition maps of an object, corresponds for a subobject to the reflection of membership along the same transitions (the two are related through quotients, Theorem 7.4). The coherent subobjects have this property, and conversely; this is a preservation theorem for intuitionistic logic, and it determines which equivalence relations on coherent objects have quotients in $\mathsf{HP}\mathcal{C}$.

Definition 7.1. Let $U \leq \eta A$ be a subobject in $\mathsf{H}\mathcal{C}$, A an object of $\mathcal{C}$. For a Kripke model M over $\mathcal{K}$ write $U_M := M^{\mathsf{H}}(U) \subseteq M(A)$. We say that U is *rigid* if for every Kripke model M, every arrow $w\colon k \to l$ of $\mathcal{K}$ such that $M(w)$ is an isomorphism of models, and every $x \in A(M_k)$,

$$M(w)_A(x) \in U_M(l) \implies x \in U_M(k).$$

The converse implication always holds, U_M being a subfunctor. For every subobject $S \leq A$ of $\mathcal{C}$ the subobject ηS is rigid, since $(\eta S)_M(k) = S(M_k)$.

Lemma 7.2. *Let $\mathcal{K}_0$ be a full subcategory of a small category $\mathcal{K}$ such that every arrow of $\mathcal{K}$ with domain in $\mathcal{K}_0$ has its codomain in $\mathcal{K}_0$. Then the restriction functor $\mathbf{Set}^{\mathcal{K}} \to \mathbf{Set}^{\mathcal{K}_0}$ is a Heyting functor. Consequently, for a Kripke model M over $\mathcal{K}$ with restriction M_0 to $\mathcal{K}_0$, and $U \leq \eta A$ in $\mathsf{H}\mathcal{C}$, one has $U_{M_0}(k) = U_M(k)$ for every object k of $\mathcal{K}_0$.*

Proof. Restriction preserves limits and colimits, hence is coherent. For $f\colon A \to B$ in $\mathbf{Set}^{\mathcal{K}}$ and a subfunctor $U \subseteq A$,

$$(\forall_f U)(k) = \{y \in B(k) \mid x \in U(l) \text{ for all } w\colon k \to l \\ \text{and all } x \in A(l) \text{ with } f_l(x) = B(w)y\},$$

which involves only the arrows out of k; so restriction preserves $\forall_f$. The restriction of M^{H} is then a Heyting functor extending M_0, hence isomorphic to M_0^{H}. □

We use the completeness theorem for Kripke models in the following form, due to Joyal: for every small Heyting category $\mathcal{D}$ there are a small category $\mathcal{K}$ and a conservative Heyting functor $\mathcal{D} \to \mathbf{Set}^{\mathcal{K}}$ [3]. For $\mathcal{D} = \mathsf{H}\mathcal{C}$ such a functor is M^{H} for the Kripke model M obtained by restriction to $\mathcal{C}$; so two subobjects $U, U' \leq \eta A$ with $U_M = U'_M$ for all Kripke models M of $\mathcal{C}$ are equal.

Theorem 7.3 (rigid formulas are coherent)**.** *Let $\mathcal{C}$ be a small coherent category and $U \leq \eta A$ a subobject in $\mathsf{H}\mathcal{C}$. Then U is rigid if and only if $U = \eta S$ for a subobject $S \leq A$ of $\mathcal{C}$.*

Proof. Let U be rigid.

Step 1: $U_M(k)$ depends only on the model M_k. Let M, M' be Kripke models over $\mathcal{K}$, $\mathcal{K}'$, let k, k' be objects of $\mathcal{K}$, $\mathcal{K}'$ and $\theta\colon M_k \to M'_{k'}$ an

isomorphism of models; replacing $\mathcal{K}'$ by a copy, we may assume that $\mathcal{K}$ and $\mathcal{K}'$ are disjoint. Let $\mathcal{K}''$ be the category whose objects are those of $\mathcal{K}$, those of $\mathcal{K}'$ and a new object o, and whose arrows are those of $\mathcal{K}$ and $\mathcal{K}'$, the identity of o, an arrow $\bar{f}\colon o \to j$ for every arrow $f\colon k \to j$ of $\mathcal{K}$, and an arrow $\bar{f}'\colon o \to j'$ for every arrow $f'\colon k' \to j'$ of $\mathcal{K}'$, with $g\bar{f} = \overline{gf}$. Let M'' be the Kripke model over $\mathcal{K}''$ which is M on $\mathcal{K}$, M' on $\mathcal{K}'$, with $M''_o = M_k$, $M''(\bar{f}) = M(f)$ and $M''(\bar{f}') = M'(f')\theta$. By Lemma 7.2, applied to $\mathcal{K}$ and to $\mathcal{K}'$, $U_{M''}$ agrees with U_M on $\mathcal{K}$ and with $U_{M'}$ on $\mathcal{K}'$. Let $x \in U_M(k)$. The arrow $\overline{\mathrm{id}_k}\colon o \to k$ acts by the identity of M_k, so $x \in U_{M''}(o)$ by rigidity; and the arrow $\overline{\mathrm{id}_{k'}}\colon o \to k'$ acts by θ, so $\theta_A(x) \in U_{M''}(k') = U_{M'}(k')$. By symmetry, $x \in U_M(k)$ iff $\theta_A(x) \in U_{M'}(k')$.

Step 2: a class of models preserved by homomorphisms. For a model N of $\mathcal{C}$, a Kripke model over the terminal category, put $U[N] := U_N \subseteq A(N)$; it is the interpretation in N of a first-order formula $\varphi_U(x)$ of the language of the classical theory $T_{\mathcal{C}}$ of Lemma 4.1, read classically. By Step 1, $U_M(k) = U[M_k]$ for every Kripke model M and every k. A homomorphism $h\colon N \to N'$ is a Kripke model over the category with one non-identity arrow, and U_M is a subfunctor; so $h_A(U[N]) \subseteq U[N']$.

Step 3: preservation. Let Φ be the set of the subobjects $S \leq A$ of $\mathcal{C}$ with $S(N) \subseteq U[N]$ for all models N; it is closed under finite unions. We claim that $U[N] \subseteq \bigcup_{S\in\Phi} S(N)$ for all N. Let $a \in U[N]$, and consider the theory $T_{\mathcal{C}}$ with the constants c_b and the axioms $f(c_b) = c_{f_N(b)}$ of the proof of Lemma 4.1, for the model N, and the axiom $\neg\varphi_U(c_a)$. It has no model, since a model would give a homomorphism $h\colon N \to N'$ with $h_A(a) \notin U[N']$. By compactness a finite part has no model; as in that proof it gives a subobject θ of a product $A \times \prod_i B_i$ and a point of $\theta(N)$ over a such that $S := \exists_{\pi_0}\theta \leq A$ satisfies $S(N') \subseteq U[N']$ for all models N'. So $S \in \Phi$ and $a \in S(N)$. By the claim and compactness, applied to the theory $T_{\mathcal{C}}$ with a constant c and the axioms $\varphi_U(c)$ and $\neg S(c)$, $S \in \Phi$, there is $S \in \Phi$ with $U[N] \subseteq S(N)$ for all N, hence $U[N] = S(N)$ for all N.

Step 4. For every Kripke model M and every k, $U_M(k) = U[M_k] = S(M_k) = (\eta S)_M(k)$. By the completeness theorem, $U = \eta S$. □

Theorem 7.3 is a preservation theorem: an intuitionistic formula is equivalent, intuitionistically over the theory, to a coherent formula if and only if its forcing at a node of a Kripke model is reflected along the transitions which are isomorphisms. Lemma 4.1 is the case $U = \neg\eta S$: the pseudo-complement is rigid if and only if it is coherent, that is, if and only if S is complemented (if $\neg\eta S = \eta S'$, then in every model S' is the complement of S, the intuitionistic negation being the complement in **Set**, and S' is a complement of S in $\mathcal{C}$ by completeness).

Applied to the coherent category $\mathsf{P}\mathcal{C}$, whose Kripke models are those of $\mathcal{C}$, the theorem determines the effective equivalence relations on the coherent objects of $\mathsf{HP}\mathcal{C}$.

Theorem 7.4. *Let $\mathcal{C}$ be a small coherent category, Y an object of $\mathsf{P}\mathcal{C}$ and E an equivalence relation on ηY in $\mathsf{HP}\mathcal{C}$. The following are equivalent.*

(i) *E is effective, that is, a kernel pair in $\mathsf{HP}\mathcal{C}$.*
(ii) *E is rigid, as a subobject of $\eta(Y \times Y)$.*
(iii) *$E = \eta F$ for an equivalence relation F on Y in $\mathsf{P}\mathcal{C}$.*

Proof. (iii)⇒(i): η is coherent, so ηF is the kernel pair of $\eta(Y \to Y/F)$. (i)⇒(ii): let E be the kernel pair of a morphism, hence of a cover $q\colon \eta Y \to X$. For a Kripke model M the functor M^{HP} preserves covers and kernel pairs, so $M^{\mathsf{HP}}(X)(k)$ is the quotient of $Y(M_k)$ by $E_M(k)$. If $w\colon k \to l$ acts by an isomorphism and the images of $x, x' \in Y(M_k)$ are related by $E_M(l)$, then the classes of x and x' have the same image in $M^{\mathsf{HP}}(X)(l)$, so they are equal by Lemma 3.1: $(x, x') \in E_M(k)$. (ii)⇒(iii): by Theorem 7.3 for $\mathsf{P}\mathcal{C}$, $E = \eta F$ with $F \leq Y \times Y$ in $\mathsf{P}\mathcal{C}$; and F is an equivalence relation, because η reflects the order of subobjects, the models of $\mathsf{P}\mathcal{C}$ being jointly conservative and each of them being $N^{\mathsf{H}}\eta$. □

So on the coherent imaginaries the free Heyting category has exactly the quotients which were there before: no equivalence relation which is new in intuitionistic logic has a quotient. This is analogous to the modal quasi-pretoposes of [2], where the equivalence relations required to be effective are those in a distinguished class of subobjects; we do not verify any axioms of quasi-pretoposes for $\mathsf{HP}\mathcal{C}$. For equivalence relations on an object $X \leq \eta Y$ which is not itself coherent, (i)⇒(ii) holds in the form that E is reflected along the isomorphic transitions among the pairs of points of X, with the same proof; the converse fails (Proposition 7.6).

The invariant does not characterise the image. The invariant of Lemma 3.1 compares a node with a node above it which carries the same model. The objects in the image of $c_{\mathcal{C}}$ satisfy a stronger condition, in which the models at several nodes are compared through homomorphisms which need not be transitions of the Kripke model.

Lemma 7.5 (cocone invariant)**.** *Let $X \leq \eta_{\mathsf{P}\mathcal{C}} Y$ be an object of $\mathsf{HP}\mathcal{C}$, Y in $\mathsf{P}\mathcal{C}$, let M be a Kripke model of $\mathcal{C}$ over $\mathcal{K}$, $\mathcal{K}'$ a subcategory of $\mathcal{K}$, N a model of $\mathcal{C}$, and $g_k\colon M_k \to N$, for the objects k of $\mathcal{K}'$, homomorphisms with $g_l \circ M(w) = g_k$ for every arrow $w\colon k \to l$ of $\mathcal{K}'$. Write $\bar{g}_k\colon Y(M_k) \to Y(N)$ for the extension of g_k to $\mathsf{P}\mathcal{C}$ at Y. Let $x_i \in M^{\mathsf{HP}}(X)(k_i)$, $0 \leq i \leq n$, be points at objects of $\mathcal{K}'$ such that for every $i < n$ there is an arrow of $\mathcal{K}'$ between k_i and k_{i+1}, in one of the two directions, which sends one of x_i, x_{i+1} to the other. Then $\bar{g}_{k_0}(x_0) = \bar{g}_{k_n}(x_n)$. In particular, if $k_0 = k_n$ and g_{k_0} is an isomorphism, then $x_0 = x_n$.*

Proof. $M^{\mathsf{HP}}(X)$ is a subfunctor of $M^{\mathsf{P}}(Y)$, whose value at k is $Y(M_k)$ and whose map along w is the extension of $M(w)$. The extension of homomorphisms to $\mathsf{P}\mathcal{C}$ is functorial; so $\bar{g}_l \circ M^{\mathsf{P}}(Y)(w) = \bar{g}_k$ for the arrows $w\colon k \to l$ of $\mathcal{K}'$, and $\bar{g}_{k_i}(x_i)$ does not depend on i. The extension of an isomorphism is a bijection. □

Lemma 3.1, for $k \neq l$, is the case of one arrow $w\colon k \to l$ with $M(w)$ an isomorphism, $N = M_k$, $g_k = \mathrm{id}$, $g_l = M(w)^{-1}$ and the points x, $M^{\mathsf{HP}}(X)(w)x = M^{\mathsf{HP}}(X)(w)x'$, x'. Another case: if $u\colon r \to k$ and $v\colon r \to l$ are arrows with $M(u) = g{\circ}M(v)$ for a homomorphism $g\colon M_l \to M_k$, then two points at r with the same image under v have the same image under u (take $\mathcal{K}'$ generated by v, $N = M_k$, $g_r = M(u)$, $g_l = g$).

Proposition 7.6. *Let D be the three-element chain $0 < p < 1$, so that $\mathsf{H}D$ is the free Heyting algebra on the generator p. Let*

$$a := \neg\neg p \Rightarrow p, \quad A := a \Rightarrow (p \vee \neg p), \quad B := A \Rightarrow (\neg\neg p \vee a), \quad W := \neg\neg p,$$

so that $W \leq A \wedge B$, and let $Z := (A + B)/E$ in $\mathsf{PH}D$ be the pushout of $A \leftarrow W \rightarrow B$, the subobjects A and B of 1 glued along W: with κ_A, κ_B the coprojections, $E := \Delta_{A+B} \vee (\kappa_A, \kappa_B)(W) \vee (\kappa_B, \kappa_A)(W)$.

(a) *For every Kripke model M of D over a small category, all the transition maps of $M^{\mathsf{PH}}(Z)$ are injective. In particular Z has the invariant of Lemma 3.1.*

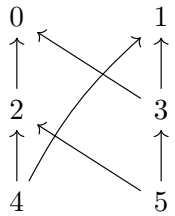
(b) *Z is not isomorphic to $c_D(X)$ for any object X of $\mathsf{HP}D$.*

(c) *The same formulas define an equivalence relation on an object $A + B \leq \eta(1 + 1)$ of $\mathsf{HP}D$ which is reflected along all transitions and is not effective.*

Proof. The models of D are the prime filters $P_0 = \{1\} \subset P_1 = \{p, 1\}$. So a Kripke model over $\mathcal{K}$ is a functor $k \mapsto |k|$ from $\mathcal{K}$ to the chain $0 < 1$; a subobject of 1 in $\mathbf{Set}^{\mathcal{K}}$ is a set of objects closed under arrows, and we write $k \Vdash U$ when k belongs to the value of $U \in \mathsf{H}D$; thus $k \Vdash p$ iff $|k| = 1$. From $\neg\neg p$ and a one gets p; so $W \leq A$, and $W \leq B$ is clear. The fibre of Z at k consists of a point s if $k \Vdash A$ and a point t if $k \Vdash B$, and $s = t$ there iff $k \Vdash \neg\neg p$.

(a) $A \wedge B \leq \neg\neg p \vee a$ and $a \wedge A \leq p \vee \neg p \leq \neg\neg p \vee \neg p$; so $A \wedge B \leq \neg\neg p \vee \neg p$. If the fibre at k has two points, then $k \Vdash A \wedge B$ and $k \nVdash \neg\neg p$, so $k \Vdash \neg p$; then no l with an arrow $k \to l$ forces $\neg\neg p$, and the two points have distinct images at l.

(b) Let M be the Kripke model over the poset with elements $0, \ldots, 5$ and order generated by the arrows shown, with $|0| = 1$ and $|k| = 0$ for $k \neq 0$:



(the arrow from 4 goes to 1). The nodes above 1, 2, 3, 4, 5 are $\{1\}$, $\{2, 0\}$, $\{3, 1, 0\}$, $\{4, 2, 1, 0\}$, $\{5, 3, 2, 1, 0\}$. One finds: $0 \Vdash p$; $1 \Vdash \neg p$, hence $1 \Vdash a$, A, B and $1 \nVdash \neg\neg p$; $2 \Vdash \neg\neg p$, hence $2 \Vdash A$, B, and $2 \nVdash a$; $3 \Vdash a$, since $\neg\neg p$ is forced only at 0 among the nodes above 3, and $3 \nVdash p \vee \neg p$, so $3 \nVdash A$; $4 \Vdash A$,

since among the nodes above 4 only 1 and 0 force a, and they force $p \vee \neg p$; $4 \nVdash \neg\neg p \vee a$, so $4 \nVdash B$; $5 \nVdash A$, since 3 is above 5; and $5 \Vdash B$, since the nodes above 5 which force A are 2, 1, 0, and they force $\neg\neg p \vee a$. So the fibres of $M^{\mathsf{PH}}(Z)$ at 4, 2, 5, 1 are $\{s\}$, $\{s = t\}$, $\{t\}$, $\{s, t\}$ with $s \neq t$ at 1.

Suppose $Z \cong c_D(X)$; then $M^{\mathsf{PH}}(Z) \cong M^{\mathsf{HP}}(X)$ by (2). Let $\mathcal{K}'$ be the full subposet on $1, 2, 4, 5$; all its nodes carry the model P_0, with identities as transitions, and $g_k = \mathrm{id}_{P_0}$ is a cocone. The points

$$s \text{ at } 1, \quad s \text{ at } 4, \quad s = t \text{ at } 2, \quad t \text{ at } 5, \quad t \text{ at } 1$$

are related successively by the arrows $4 \to 1$, $4 \to 2$, $5 \to 2$, $5 \to 1$. By Lemma 7.5, $s = t$ at 1; a contradiction.

(c) We use the same letters for the subobjects of $\eta 1$ in $\mathsf{HP}D$ defined by the same formulas. In $\mathsf{HP}D$ let $A + B$ be the subobject of $\eta(1 + 1)$ which is A on the first summand and B on the second; the formula for E defines an equivalence relation E_0 on it with $c_D(E_0) = E$. It is reflected along all transitions by (a). If E_0 were the kernel pair of a morphism, it would be the kernel pair of a cover $A + B \to X$, and c_D, a Heyting functor, preserves covers and kernel pairs; so $c_D(X) \cong Z$, contradicting (b). □

So injectivity of the transition maps, along the isomorphisms of models or even along all transitions, does not characterise the essential image of $c_{\mathcal{C}}$, already for the smallest non-Boolean propositional theory: Z is the union of two subobjects A and B which lie in the image, all its transition maps are injective, and it is excluded only by a zigzag through four nodes carrying the same model. For two copies of a coherent object ηY glued along W, on the other hand, the invariant makes W rigid, hence coherent (Theorem 7.3), and the glued object is coherent.

The image for finite propositional theories. For a finite propositional theory the cocone invariant characterises the image, and it has to be tested in one finite Kripke model only.

Lemma 7.7. *For every small coherent category $\mathcal{C}$, an object of $\mathsf{PH}\mathcal{C}$ is in the essential image of $c_{\mathcal{C}}$ if and only if it admits a monomorphism to $(\mathsf{P}\eta)(Y)$ for some object Y of $\mathsf{P}\mathcal{C}$.*

Proof. An object of $\mathsf{HP}\mathcal{C}$ is a subobject $X \leq \eta_{\mathsf{P}\mathcal{C}} Y$, and $c_{\mathcal{C}}(\eta_{\mathsf{P}\mathcal{C}} Y) = (\mathsf{P}\eta)(Y)$. Conversely let $V \leq (\mathsf{P}\eta)(Y)$, and let $q\colon \coprod_j A_j \to Y$ be a cover in $\mathsf{P}\mathcal{C}$ with the A_j in $\mathcal{C}$. The pullback of V along $(\mathsf{P}\eta)(q)$ is a family of subobjects $V_j \leq \eta A_j$, which belong to $\mathsf{H}\mathcal{C}$, the unit $\mathsf{H}\mathcal{C} \to \mathsf{PH}\mathcal{C}$ being full on subobjects [1, A3.3]. Let $\mathsf{H}\iota\colon \mathsf{H}\mathcal{C} \to \mathsf{HP}\mathcal{C}$ be the Heyting functor induced by the unit $\iota\colon \mathcal{C} \to \mathsf{P}\mathcal{C}$; then $c_{\mathcal{C}} \circ \mathsf{H}\iota$ is isomorphic to the unit $\mathsf{H}\mathcal{C} \to \mathsf{PH}\mathcal{C}$, both being Heyting functors which restrict to $(\mathsf{P}\eta)\iota$ on $\mathcal{C}$. In $\mathsf{HP}\mathcal{C}$ let $V' \leq \eta_{\mathsf{P}\mathcal{C}} Y$ be the image under $\eta_{\mathsf{P}\mathcal{C}}(q)$ of the union of the subobjects $(\mathsf{H}\iota)(V_j)$ of the summands of $\eta_{\mathsf{P}\mathcal{C}}(\coprod_j A_j)$. Then $c_{\mathcal{C}}(V')$ is the image of $\coprod_j V_j$ under the cover $(\mathsf{P}\eta)(q)$, which is V. □

Let now D be a finite distributive lattice and X the poset of its prime filters, the models of D; the homomorphisms are the inclusions, and $\mathsf{P}D$ is the category $\mathbf{FinSet}^X$ of functors from X to finite sets, the coherent objects of the classifying topos $\mathbf{Set}^X$ [1, D3.3]; in the same way $\mathsf{P}D' = \mathbf{FinSet}^{X'}$ for every finite distributive lattice D' with poset of prime filters X', an element of D' corresponding to the subterminal functor supported on the prime filters containing it. Let $H_0 \subseteq H_1 \subseteq \cdots$ be the sublattices of $\mathsf{H}D$ with $H_0 = D$ and H_{n+1} generated by D and the elements $u \Rightarrow v$ with $u, v \in H_n$. They are finite, being finitely generated distributive lattices, and their union is $\mathsf{H}D$. Let X_n be the poset of the prime filters of H_n, ordered by inclusion, and $|F| := F \cap D$; so $F \mapsto |F|$ is a Kripke model of D over X_n.

Lemma 7.8. *In the Kripke model X_n, for $U \in H_n$ and $F \in X_n$: $F \Vdash U$ if and only if $U \in F$.*

Proof. By induction on $k \leq n$ we prove it for $U \in H_k$. The filters being prime and forcing commuting with finite meets and joins, it suffices to treat the generators. For $d \in D$, $F \Vdash d$ iff $d \in |F|$. Let $u, v \in H_{k-1}$. If $u \Rightarrow v \in F$ and $F \subseteq F' \Vdash u$, then $u \in F'$ by induction, so $v \in F'$ and $F' \Vdash v$; hence $F \Vdash u \Rightarrow v$. If $u \Rightarrow v \notin F$, let $\hat{F}$ be a prime filter of $\mathsf{H}D$ with $\hat{F} \cap H_n = F$; it exists because the filter of $\mathsf{H}D$ generated by F and the ideal generated by $H_n \setminus F$ are disjoint. The filter generated by $\hat{F}$ and u does not contain v, since $f \wedge u \leq v$ with $f \in \hat{F}$ gives $u \Rightarrow v \in \hat{F} \cap H_n$. So there is a prime filter $\hat{G} \supseteq \hat{F}$ of $\mathsf{H}D$ with $u \in \hat{G}$, $v \notin \hat{G}$, and $F' := \hat{G} \cap H_n \supseteq F$ has $F' \Vdash u$ and $F' \nVdash v$ by induction. Hence $F \nVdash u \Rightarrow v$. □

Every object of $\mathsf{PH}D$ is a quotient $Z = (\coprod_{i \in I} A_i)/E$ of a finite coproduct of elements A_i of $\mathsf{H}D$ by an equivalence relation, which is given by elements $E_{ij} \leq A_i \wedge A_j$; the fibre of Z at a node k of a Kripke model is the set of the i with $k \Vdash A_i$ modulo the relation $k \Vdash E_{ij}$.

Theorem 7.9. *Let D be a finite distributive lattice, $Z = (\coprod_i A_i)/E$ an object of $\mathsf{PH}D$, and n such that all A_i and E_{ij} belong to H_n. The following are equivalent.*

(i) *Z is in the essential image of c_D.*
(ii) *For every Kripke model M of D over a poset $\mathcal{K}$ and every node k: two points of the fibre of Z at k which are connected by a zigzag, in the sense of Lemma 7.5, through nodes l with $M_l \subseteq M_k$, are equal.*
(iii) *Condition (ii) for the finite Kripke model X_n.*

If they hold, Z is a subobject of $(\mathsf{P}\eta)(Y)$ for the functor Y on X with $Y(x) = \operatorname{colim}_{|F| \subseteq x} Z(F)$, the left Kan extension along $F \mapsto |F|$ of the fibres of Z on X_n.

Proof. (i)⇒(ii): if $Z \cong c_D(X')$, then $M^{\mathsf{PH}}(Z) \cong M^{\mathsf{HP}}(X')$, and Lemma 7.5 applies to the full subposet $\mathcal{K}'$ of the nodes l with $M_l \subseteq M_k$, the model $N = M_k$ and the inclusions $g_l \colon M_l \subseteq M_k$, with $g_k = \mathrm{id}$. (ii)⇒(iii) is clear.

(iii)⇒(i). The inclusions $D \subseteq H_n \subseteq \mathsf{H}D$ are homomorphisms of lattices and induce coherent functors

$$\mathsf{P}D = \mathbf{FinSet}^X \longrightarrow \mathsf{P}H_n = \mathbf{FinSet}^{X_n} \longrightarrow \mathsf{PH}D$$

with composite $\mathsf{P}\eta$, the first being composition with $F \mapsto |F|$. The elements E_{ij} define an equivalence relation on $\coprod_i A_i$ in $\mathsf{P}H_n$, the conditions being inequalities between elements of H_n which hold in $\mathsf{H}D$; let Z_n be the quotient, an object of $\mathsf{P}H_n$ whose image in $\mathsf{PH}D$ is Z. As a functor on X_n, $Z_n(F)$ is the set of the i with $A_i \in F$ modulo the relation $E_{ij} \in F$; by Lemma 7.8 this is the fibre of Z at F in the Kripke model X_n. Let Y be the left Kan extension of Z_n along $F \mapsto |F|$, a functor from X to finite sets: $Y(x)$ is the colimit of $Z_n(F)$ over the F with $|F| \subseteq x$, the set of the pairs (F, z), $z \in Z_n(F)$, modulo the equivalence relation generated by $(F, z) \sim (F', Z_n(F \subseteq F')z)$. The unit $Z_n(F) \to Y(|F|)$, $z \mapsto [(F, z)]$, is natural in F, and it is injective if and only if two points of $Z_n(F)$ connected by a zigzag through filters F' with $|F'| \subseteq |F|$ are equal, which is (iii). So the unit is a monomorphism from Z_n to the image of Y in $\mathsf{P}H_n$; applying the coherent functor to $\mathsf{PH}D$ gives a monomorphism $Z \to (\mathsf{P}\eta)(Y)$, and Lemma 7.7 applies. □

So for a finite propositional theory the essential image of c_D consists of the objects with the cocone invariant, and the embedding into a coherent object, when it exists, is the canonical one computed in the finite model X_n. Given the finite lattice D and formulas for the A_i and E_{ij}, of implication depth at most n, the lattice H_n and the poset X_n can be computed, intuitionistic propositional logic being decidable; so membership in the image is decidable. The object Z of Proposition 7.6 fails (ii) in the six-element model given there. Theorem 7.9 uses the finiteness of D through the finite models X_n, in which every element of H_n is forced exactly where it should be; we do not know a substitute for first-order theories (Question 9.2).

8. When excluded middle repairs the comparison

Let $\mathsf{K}\colon \mathbf{Heyt} \to \mathbf{BCoh}$ be the classicalisation, the reflector onto the full sub-2-category **BCoh** of Boolean coherent categories which adds excluded middle to the internal first-order theory; $\mathsf{B} \simeq \mathsf{KH}$ is the Morleyization, and $\mathsf{KP}\mathcal{D} \simeq \mathsf{PK}\mathcal{D}$ for every Heyting category $\mathcal{D}$ [5, Thm. 9.2]. The comparison $b_{\mathcal{C}}\colon \mathsf{BP}\mathcal{C} \to \mathsf{PB}\mathcal{C}$ of [5] is $\mathsf{K}c_{\mathcal{C}}$ followed by this equivalence at $\mathcal{D} = \mathsf{H}\mathcal{C}$ [5, Cor. 9.3]:

$$\begin{array}{ccccc}
\mathsf{HP}\mathcal{C} & & \xrightarrow{\quad c_{\mathcal{C}} \quad} & & \mathsf{PH}\mathcal{C} \\
\downarrow & & & & \downarrow \\
\mathsf{BP}\mathcal{C} = \mathsf{KHP}\mathcal{C} & & \xrightarrow{\mathsf{K}c_{\mathcal{C}}} & \mathsf{KPH}\mathcal{C} & \xrightarrow{\quad \simeq \quad} \mathsf{PKH}\mathcal{C} = \mathsf{PB}\mathcal{C}.
\end{array}$$

$$\mathsf{BP}\mathcal{C} \xrightarrow{\quad b_{\mathcal{C}} \quad} \mathsf{PKH}\mathcal{C}$$

Corollary 8.1. *For a small coherent category $\mathcal{C}$ the following are equivalent: the Morleyization commutes with the pretopos completion at $\mathcal{C}$, that is*

$b_{\mathcal{C}}$ is an equivalence; the Galois criterion of [5, Thm. 6.1] *holds for $\mathcal{C}$ (in the saturated, strongly homogeneous models of the completions of the classical theory, every element of a classical imaginary sort has the same automorphism stabiliser as some element of a coherent imaginary sort of the same model); the Heyting functor $c_{\mathcal{C}}$ is inverted by K. If these conditions hold and $\mathcal{C}$ is not Boolean, then $c_{\mathcal{C}}$ is a full and faithful Heyting functor which is not an equivalence and becomes one when excluded middle is added. This is the case for the theory of pure equality, for the theory of an equivalence relation, for the theories without axioms over a monadic relational signature with at least one sort, and for the non-Boolean propositional theories* [5, Ex. 6.5, Thm. 8.2, Thm. 7.1, Prop. 5.2].

Proof. The first equivalence is [5, Thm. 6.1], the second the factorisation above, and the rest is Theorem 1.1. The theories listed are not Boolean: in each there is a homomorphism of models which is not an elementary embedding (a non-injective map of sets in the first and third cases, a map merging two classes, a proper inclusion of prime filters). □

In the example of pure equality one sees the repair: classically $\neg(x = y)$ is a complemented subobject of $\sigma \times \sigma$ and the glued object Q of Example 5.2 becomes the coproduct of $\sigma \times \sigma$ and the diagonal, a classical subobject of a coherent imaginary. For the theory of complete multipartite graphs [5, Thm. 5.3] the repair fails: K does not invert $c_{\mathcal{C}}$. Thus the statement of [5] that the forced order of the Morleyization and the completion arises in the Heyting completion is to be read as follows. The comparison is never an equivalence before excluded middle is added, except trivially; the Galois criterion decides whether excluded middle repairs it.

9. Questions

Question 9.1. Let $\mathcal{C}$ be a non-Boolean coherent category such that, whenever a homomorphism $h\colon M \to N$ crosses a subobject, every object of $\mathcal{C}$ has at most one point in M (for instance the product of a non-Boolean bounded distributive lattice and a Boolean coherent category). Is $\mathsf{HE}\mathcal{C}$ effective?

Question 9.2. Describe the essential image of $c_{\mathcal{C}}$ in $\mathsf{PH}\mathcal{C}$. The invariant of Lemma 3.1 is necessary, and by Theorem 7.4 it is sufficient for the quotients of coherent objects, but not in general (Proposition 7.6). The cocone invariant of Lemma 7.5 is necessary and, for finite propositional theories, sufficient (Theorem 7.9). Is it sufficient for all coherent categories, for instance for the theory of pure equality?

Question 9.3. Let L be an intermediate predicate logic, given by axiom schemes added to intuitionistic first-order logic, and H_L the corresponding

translation. The fork is not a frame for the logic of linear frames (Gödel–Dummett), and over a finite linear order two consecutive nodes whose transition is an isomorphism of models force the same formulas, under this isomorphism; so the proof of Theorem 1.1 does not apply. For which L does H_L commute with P beyond the Boolean case?

Question 9.4. Is there a description of the class of coherent functors inverted by H, and of its relation to the classes V_{B} and V_{P} of [5] in the lattice of two-out-of-three classes, in spite of H not being a reflector?

Declarations

The author has no competing interests to declare.

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